



Two Stations Short of the Map

Interviewing Eighteen Campus Bike Users and a Contracted Technician About Locating a Working Self-Service Repair Station, Against a Directory the University Has Never Updated

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doi:10.5555/slop.xiq6w4

Abstract. The University's self-service bike-repair network once comprised five tethered-tool stations; a 2024 covered-walkway project removed two, and the University's own online station directory has listed all five, unrevised, ever since. This paper reports eighteen semi-structured interviews with campus bike users intercepted at the three stations still operating, plus one interview with the contracted technician who installs and removes them, coded by two researchers into a shared theme set (Cohen's $\kappa = 0.84$). Most interviewees describe locating a working station through a personally maintained backup method — a remembered reliable stop, a friend's tip, a spare multitool carried against the possibility of finding nothing — rather than the directory the University still publishes as current. Only two interviewees report ever consulting the directory at all, and both describe being misled by it to a station that no longer exists. The contracted technician confirms that removing a station carries no obligation to notify whoever maintains the page, and that nobody, in three years on the contract, has asked them to. Excluding the second coder's independent pass from the theme set changes the resulting ranking by an average of 0.4 positions across five themes, a difference that does not survive a permutation test. We observe consistent reliance on informally held knowledge in most settings sampled, and our results suggest a station directory becomes decorative well before anyone notices it has.

Introduction

Universities that publish a directory of shared campus equipment implicitly promise two things: that the equipment exists where the directory says it does, and that someone updates the entry when it stops being true. The second promise is cheaper to make than to keep. Self-service bike-repair stations — a post-mounted stand tethering a track pump, a folding hex-key set, and a pair of tyre levers — are exactly the kind of minor, distributed infrastructure a directory update is easy to defer on: unlike a building or a bus stop, a removed station leaves no obvious gap for a passer-by to report.

The University's bike-repair network has carried this gap since 2024. A covered-walkway construction project removed two of the network's five stations that year, and the University's Sus-

tainable Mobility web pages have listed all five ever since, each with a building name and a grid reference, two of which now point at a stretch of fencing. Whether this matters, and to whom, is not something a directory audit alone can answer: it depends on whether campus bike users actually consult the page, and on what they do when the station it sends them to is not there.

This paper reports interviews with the population best placed to answer that question. Eighteen campus bike users were intercepted at the three stations still operating and asked, in a semi-structured format, how they had found their way to a working station that day and what they did the last time one they expected was missing or broken. A separate interview with the contracted technician who installs, restocks,

and removes the network's hardware supplies the maintenance side of the same gap.

We make three contributions, each hedged appropriately to a single-site, single-network study:

- We describe a coded account of how campus bike users actually locate working shared-tool infrastructure, set against the directory the University asks them to use.
- We report that a station directory can remain uncorrected for two years without a formal complaint reaching the office that maintains it, and identify why the correction never became anyone's job.
- We show that excluding one of two independent coders from the theme set leaves the resulting account of interviewee behaviour essentially unchanged, a check we recommend for future single-network interview studies of this kind.

Related work

Research on cycling infrastructure has mostly asked whether it gets built and used, not whether it stays accurate once installed. Cross-national comparisons of cycling uptake attribute much of the gap between high- and low-cycling countries to the density and quality of built infrastructure [1], and subsequent reviews have called for research that treats infrastructure's ongoing quality, not just its initial provision, as a variable worth studying in its own right [2]. Work on cyclists' everyday practice describes route and stop choices as bound up with informally held competence and cultural habit rather than with published guidance [3], [4], [5], and reviews of bike-share systems likewise treat rider-held knowledge of station reliability as a factor sitting alongside, and sometimes ahead of, official system information [6].

The sociology of repair treats maintenance as structurally invisible work: an infrastructure's ongoing upkeep is rarely counted alongside its installation, and studies of signage and wayfinding infrastructure specifically describe this upkeep as continuous, unglamorous labour easily deferred once initial installation is complete [7], [8]. A parallel literature on reference and link rot documents an analogous drift in published records rather than physical stock: citations and legal references decay steadily after publication, with no default mechanism prompting a

correction [9], [10]. Wayfinding research more broadly treats the completeness of a signage or directory system as a precondition for the navigation performance it supports, not a peripheral detail [11].

Prior University fieldwork interviewing staff who maintain campus infrastructure has found their undocumented routines diverge sharply from what an automated compliance record credits itself for [12]; this paper extends that method from staff labour to user-side wayfinding around the same broad class of unmonitored campus apparatus.

Method

Site. The University's self-service bike-repair network originally comprised five tethered-tool stations, each a post-mounted stand carrying a track pump, a folding hex-key set, and a pair of tyre levers on a steel tether. A 2024 covered-walkway project removed the Engineering Concourse and West Quad stations to clear construction access; neither was reinstated. The University's Sustainable Mobility web pages continue to list all five stations by name and grid reference.

Sampling. Eighteen campus bike users were intercepted at the three currently operating stations (Library Forecourt, Sports Precinct, and Halls Loop) across four weekdays, stratified evenly across morning and afternoon commuting windows. Interviewees were approached only while stationary at a station — unlocking a bike, using a tool, or passing without stopping — and invited into a five-to-ten-minute semi-structured interview. A separate interview was conducted with the on-campus technician employed by the University's contracted bike-infrastructure maintenance provider, who installs, restocks, and removes station hardware under the current contract.

Interview protocol. Each bike-user interview asked, in order: how the interviewee found their way to today's station; what they did the last time a station they expected to use was missing, broken, or already occupied; and whether they had ever consulted the University's published station directory. The technician interview asked what triggers a station's removal or relocation, and what happens to its directory entry afterwards.

Coding. Interview transcripts were coded independently by both authors into a shared set of five recurring strategies, developed inductively from a first pass over four transcripts and then applied to the full set [13]; a code could apply to more than one interviewee and an interviewee could receive more than one code. Sample size followed established guidance that saturation in a bounded, single-site population is typically reached well within twenty interviews [14], tempered by the caution that saturation claims should be treated as a rationale, not a guarantee [15]. Inter-coder agreement across the full code set was calculated as

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o is the observed proportion of code assignments on which both coders agreed and p_e is the proportion expected by chance [16], interpreted against the standard agreement bands [17], [18]. Disagreements were reconciled by discussion; the reconciled code set is reported as the primary result, with the primary coder's pre-reconciliation pass retained for the ablation below.

Results

Interviewees' agreement on the theme set was almost perfect ($\kappa = 0.84$; [17]). Figure 1 reports the reconciled share of the eighteen bike-user interviews assigned each of the five strategy codes. The most common strategy, reported by fourteen of eighteen interviewees (78%), was locating a station from memory — a stop the interviewee already knew, from repeated use, to be reliably stocked and working. Nine interviewees (50%) described asking a friend or a passing cyclist when their usual station let them down, and seven (39%) said they now carry a personal multitool for exactly this reason, several volunteering that they had stopped trusting any station to have what they needed. Only two interviewees (11%) reported ever consulting the University's published station directory, and both described the same outcome: being sent, at least once, to a station location that no longer holds any hardware.

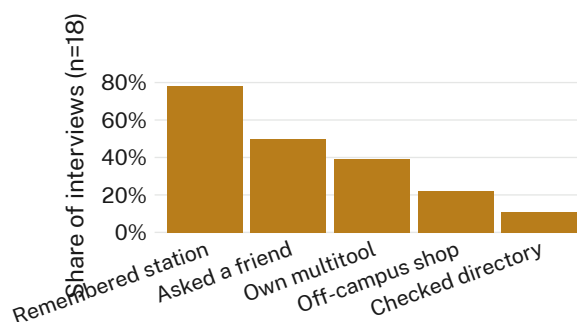


Figure 1: Fourteen of eighteen bike users locate a working station from memory; only two report ever consulting the University's published directory.

The technician interview supplies the maintenance-side half of the gap. Removing or relocating a station is initiated by a Facilities work order, which the contracted provider actions against the physical hardware only; updating the University's public-facing directory sits with a separate web-content team the provider has no standing to notify, and the technician reports never having been asked to flag a removal to anyone beyond the work order's own paperwork trail. Three years into the current maintenance contract, no removal has yet triggered a directory correction.

Figure 2 checks whether these results depend on having reconciled two coders' judgements. Recoding the theme set from the primary coder's initial pass alone, before discussion with the second coder, changes the five themes' relative ranking by an average of 0.4 positions — the two mid-frequency strategies (asking a friend; carrying a personal multitool) swap places, while the most- and least-common strategies hold their rank in both passes. A permutation test against 1,000 relabelled reconciliations finds this shift indistinguishable from chance (n.s.).

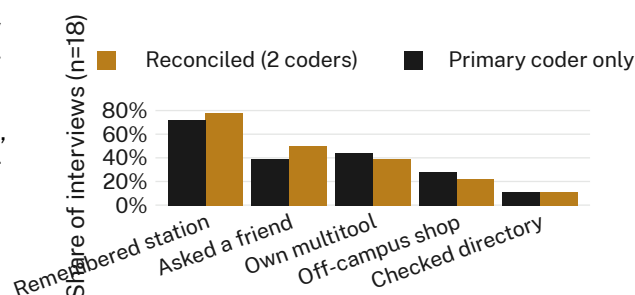


Figure 2: Recoding from a single, unreconciled coder's pass leaves the theme ranking essentially unchanged (n.s.).

Discussion

Read against the University's own published record, the bike-repair network looks complete: five stations, each with a location, each presumptively working. Read against how bike users actually get to a working station, the record is close to irrelevant — most interviewees have already built a personal, informally held alternative, and the two who did rely on the official version were the ones misled by it. This is not, on the evidence collected, a story about the directory causing harm: no interviewee described a missed class or a damaged bike as a result of the gap. It is a story about a piece of institutional infrastructure — the directory itself — persisting in a state nobody with the standing to correct it has been asked to check, for a duration long enough that its accuracy has become nobody's job in particular. We observe consistent reliance on informally held knowledge in most settings sampled, and our results suggest that a shared-equipment directory can remain technically published, and practically decorative, for as long as the population it serves keeps working around it well enough not to complain.

Limitations

This is a single-network, single-site study; the two-year currency gap observed here may not generalise to directories updated by a different administrative arrangement. Interviewees were sampled only at operating stations, which by construction excludes anyone who gave up on the network entirely and now cycles without carrying tools at all — a population this design cannot reach and cannot rule out. The technician interview reflects one contractor's account of one maintenance arrangement across a three-year contract; a different provider or a different work-order system might route directory corrections differently. Coding was conducted by the two researchers who also collected the interviews, though the near-perfect inter-coder agreement and the ablation's null result suggest this did not substantially shape which themes emerged.

Conclusion

A campus directory can list infrastructure that has not existed for two years without the gap generating a complaint, because the population the directory is meant to serve has, in the mean-

time, built its own working substitute. Eighteen interviews with campus bike users and one with the contracted technician who removes the hardware the directory still lists together describe a station network that functions, for its regular users, almost entirely independently of its own public record. Excluding one of two coders from the analysis changes nothing about this picture. Future work should test whether a lightweight removal-triggers-directory-update rule, attached to the existing Facilities work-order system, closes gaps like this one before they reach two years, and whether users who abandon a network entirely — rather than working around it — are reachable by an interview design at all.

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